# Optimal Reconfiguration of Systems in Aeronautics

## Lecture 05: "Introduction to Linear Optimal Control Systems"

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# Outline


1. Nominal and Fault Tolerant Linear Quadratic Control
2. B747 Application Example

# Objective I

This lecture provides a fault tolerant perspective on the linear quadratic control technique.

# Outline


1. Nominal and Fault Tolerant Linear Quadratic Control
   - Introduction
   - Nominal Operation
   - Faulty Operation
   - Fault Tolerant Operation
2. B747 Application Example

# Introduction I

Properties of interest in the analysis of the nominal and of the post-fault systems too range from stability to optimality.

Analyzing optimality can be done using the Linear Quadratic (LQ) control technique that allows the designer to take into account both requirements on the settling time of state variables and the amplitude of control inputs; optimality can also be expressed in terms of the minimum time and/or the minimum energy needed by the system transfer from one state to another.

Regarding the post-fault behavior, an automatic control system is said to be Linear Quadratic Fault Tolerant (LQ-FT) as it provides that a quadratic objective can still be satisfied in the presence of faults belonging to a given set.

# Introduction II

Unfortunately, while the Fault Detection, Isolation, and Identification / Estimation (FDIE) procedures provide information about the new system parameters / structure and the Fault Accommodation and Reconfiguration (FAR) procedures redesign / reconfigure the controller accordingly, there remains a period during which the faulty system is controlled using the nominal controller. This may possibly lead to system instability and/or control inadmissibility w.r.t. the fault tolerance objectives.

In both cases, respectively nominal and faulty, the optimal control law of a system is computed upon the solution of a matrix continuous algebraic Riccati equation (CARE). In order to shorten the impairment augmentation of the post-fault system, a successive application of CARE solutions provided at each step by an iterative technique is recommended and expected to improve the closed-loop behavior.

To this end, two iterative techniques for the accommodation of the post-fault controller are approached, namely Newton-Kleinman and Matrix-Sign-Function, and the initial step problems are tackled, as well. The benefits brought by using the progressive fault accommodation are illustrated along with the development of an application to the short-period mode control of the longitudinal motion of an aircraft.

# Nominal operation I

Consider the continuous linear time-invariant (LTI) deterministic system whose nominal operation is modeled by

$$M_n : \quad \dot{x}(t) = A_n x(t) + B_n u_n(t) \,, \text{ with } x(t_0) = x_0 \,, \tag{1}$$

where $A_n \in \mathbb{R}^{n \times n}$ and $B_n \in \mathbb{R}^{n \times m}$ are the nominal system and control matrices, $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ are the state and control vectors, and $x_0 \in \mathbb{R}^n$ is the initial condition.

Define the quadratic cost functional as

$$J[x_0, u(t)] = \int_0^{+\infty} \left[ x^\top(t) Q_n x(t) + u^\top(t) R_n u(t) \right] dt \,, \tag{2}$$

where $Q_n = \bar{Q}_n^\top \bar{Q}_n \geq 0$ and $R_n = R_n^\top > 0$ are weighting symmetric matrices, and $(.)^\top$ stands for transposition.

# Nominal operation II

Under the assumptions of stabilizability and detectability of pairs $(A_n, B_n)$ and $(\bar{Q}_n, A_n)$, respectively, the nominal linear state-feedback control law

$$u_n(t) = -K_n x(t) \tag{3}$$

is optimal for

$$K_n = R_n^{-1} B_n^\top X_n \,, \tag{4}$$

where matrix $X_n$ has to be determined.

In (4), matrix $X_n \in \mathbb{R}^{n \times n}$ is the unique stabilizing symmetric positive-semidefinite solution ($X_n^\top = X_n \geq 0$) of the nominal continuous algebraic Riccati equation

$$CARE_n : \;\; A_n^\top X_n + X_n A_n - X_n B_n R_n^{-1} B_n^\top X_n + Q_n = 0 \,. \tag{5}$$

Then, in the nominal situation, the closed-loop system

$$\hat{M}_n : \;\; \dot{x}(t) = \hat{A}_n x(t) \,, \;\; \text{with} \;\; \hat{A}_n = A_n - B_n K_n \,, \tag{6}$$

is stable, where $\hat{A}_n \in \mathbb{R}^{n \times n}$ denotes the nominal closed-loop matrix.

# Nominal operation III

Moreover, provided the solution $X_n$ to CARE$_n$ (5), the cost functional (2) takes the minimum value

$$J_{M_n}(x_0, u_n) \triangleq J_n = x_0^\top X_n x_0 \ (\equiv J^*)\,; \tag{7}$$

still, in view of the next development, the value of the optimal cost $J^*$ in (7) needs special attention.

Successively, construct the quadratic form associated with the solution $X_n$ to CARE$_n$, i.e.,

$$V_n\left[x(t)\right] = x^\top(t) X_n x(t)\,,$$

do its derivation w.r.t. time, that is

$$\begin{aligned}\frac{d}{dt}\left[x^\top(t) X_n x(t)\right] =& \dot{x}^\top(t) X_n x(t) + x^\top(t) X_n \dot{x}(t)\\ =& \left[A_n x(t) + B_n u_n(t)\right]^\top X_n x(t)\\ &+ x^\top(t) X_n \left[A_n x(t) + B_n u_n(t)\right]\,,\end{aligned}$$

# Nominal operation IV

and, using (5), obtain

$$\begin{aligned}\frac{d}{dt}\left[x^\top(t)X_n x(t)\right] =& x^\top(t)\left(A_n^\top X_n + X_n A_n\right)x(t)\\ &+ u_n^\top(t)B_n^\top X_n x(t) + x^\top(t)X_n B_n u_n(t)\\ =& x^\top(t)\left(X_n B_n R_n^{-1} B_n^\top X_n - Q_n\right)x(t)\\ &+ u_n^\top(t)B_n^\top X_n x(t) + x^\top(t)X_n B_n u_n(t)\\ =& \left[u^\top(t) + x^\top(t)X_n B_n R_n^{-1}\right]R_n\left[u_n(t) + R_n^{-1}B_n^\top X_n x(t)\right]\\ &- \left[x^\top(t)Q_n x(t) + u_n^\top(t)R_n u_n(t)\right],\end{aligned}$$

or

$$\begin{aligned}x^\top(t)Q_n x(t) + u_n^\top(t)R_n u_n(t) =& -\frac{d}{dt}\left[x^\top(t)X_n x(t)\right]\\ &+ \left[u_n^\top(t) + x^\top(t)X_n B_n R_n^{-1}\right]R_n\\ &\cdot\left[u_n(t) + R_n^{-1}B_n^\top X_n x(t)\right];\end{aligned}$$

# Nominal operation V

further, integrate it w.r.t. time, from $0$ to $T$, that is

$$\int_0^T \left[x^\top(t)Q_n x(t) + u_n^\top(t)R_n u_n(t)\right] dt = x_0^\top X_n x_0 - x^\top(T)X_n x(T) + \int_0^T \left[u_n^\top(t) + x^\top(t)X_n B_n R_n^{-1}\right] R_n \cdot \left[u_n(t) + R_n^{-1}B_n^\top X_n x(t)\right] dt\,;$$

moreover, the second term vanishes as $T \to +\infty$ and $x(T)_{T\to+\infty} \to 0$, hence giving

$$\lim_{T\to+\infty} J_n = x_0^\top X_n x_0 + \lim_{T\to+\infty} \int_0^T \left[u_n^\top(t) + x^\top(t)X_n B_n R_n^{-1}\right] R_n \cdot \left[u_n(t) + R_n^{-1}B_n^\top X_n x(t)\right] dt\,;$$

# Nominal operation VI

besides, as $R_n$ is symmetric and positive-definite, it follows that $J_n \geq x_0^\top X_n x_0$, for all initial conditions, $x_0$, and for all controls, $u_n(t)$; however, since the first term is independent of $u_n(t)$, the minimum of $J_n$ is attained for $u_n(t) = -R_n^{-1} B_n^\top X_n x(t)$; therefore, the optimal value of $J_n$ goes to $x_0^\top X_n x_0$, which is exactly $J^*$.

# Faulty Operation I

Whether parametric faults are considered, the system to be controlled is described by the pair $(A_f, B_f) \neq (A_n, B_n)$, assuming that an LTI model is still able to capture the system behavior.

There are two different means by which the fault can be recovered according to the Fault Tolerance (FT) strategy used, namely the accommodation and the reconfiguration approaches:

(i) In the accommodation strategy, the new control applies to the faulty system; its model $(A_f, B_f)$ must be identified online by the diagnostic algorithm, in addition to its fault detection and isolation functionalities.

(ii) In the reconfiguration strategy, faulty components are switched off and only the healthy part of the system is controlled; its model $(A_f, B_f)$ is therefore easily known, after fault detection and isolation has been performed.

Whatever the selected fault recovery technique, the best strategy consists of applying the optimal control solution to the system

$$M_f: \quad \dot{x}(t) = A_f x(t) + B_f u_f(t)\,, \text{ with } x_{t_f} = x(t_f) \triangleq x_f\,, \tag{8}$$

on the post-fault time interval, i.e., $t \in [t_f, +\infty)$.

# Faulty Operation II

At any rate, finding the optimal control associated with the faulty system is a controller redesign problem; it therefore amounts to solving the CARE associated with the new situation, i.e.,

$$CARE_f: \quad A_f^\top X_f + X_f A_f - X_f B_f R_n^{-1} B_f^\top X_f + Q_n = 0\,. \tag{9}$$

Assuming that the pairs $(A_f, B_f)$ and $(\bar{Q}_n, A_f)$ are still stabilizable respectively detectable, the post-fault control law calculates as

$$u_f(t) = -K_f x(t)\,, \tag{10}$$

with

$$K_f = R_n^{-1} B_f^\top X_f\,. \tag{11}$$

Forth, similarly to (6), the post-fault closed-loop system behavior is modeled by

$$\hat{M}_f: \quad \dot{x}(t) = \hat{A}_f x(t)\,, \text{ with } \hat{A}_f = A_f - B_f K_f\,, \tag{12}$$

where $\hat{A}_f$ is the post-fault closed-loop matrix.

# Faulty Operation III

Then, the optimal post-fault control law (10) leads the cost functional (2) to the value

$$J_{M_f}(x_f, u_f) \triangleq J_f = x_0^\top X_n x_0 + x_f^\top (X_f - X_n) x_f \,. \tag{13}$$

This far, (1)-(7) and (8)-(13) define the Fault Tolerant Control (FTC) problem in the LQ framework (LQ-FTC).

# Fault tolerant operation I

However, the fact that the solution $u_f(t)$ achieves the objective $J_f$ optimally does not mean that it is satisfactory. Indeed, the cost $J_f$ might exceed some given upper limit $J_{\max} \geq J^*$, thus denying system $M_n$ to deserve the "fault tolerant" label w.r.t. the fault situation $M_f$.

Next, the concept of admissibility of an optimal fault tolerant state-feedback control solution is introduced.

Definition (Admissibility)

Assuming that a solution $(u_n, u_f)$ to the LQ-FTC problem (1)-(13) exists, for all initial conditions $x_0$, it is admissible if and only if

$$J_f \leq \rho J^* \,, \tag{14}$$

where (14) defines the maximal performance degradation admitted in faulty situations, with $\rho = \frac{J_{\max}}{J^*} \geq 1$. $\square$

In this perspective, system (1) is fault tolerant w.r.t. the fault situation (8) if and only if either the accommodation or the reconfiguration problem has an admissible solution.

# Fault tolerant operation II

Naturally, (13) provides the condition to testing whether the fault $(A_f, B_f)$ is recoverable or not, but see however that both this test and the determination of the accommodated control need the $\text{CARE}_f$ to be solved.

Therefore, the interest turns on the methods that are available to obtain the solution of a continuous-time algebraic matrix Riccati equation.

In what follows, Schur respectively Generalized Schur and Eigenvector Methods as direct approaches to solving $\text{CARE}_f$ are investigated.

# Outline


1. Nominal and Fault Tolerant Linear Quadratic Control
2. B747 Application Example
   - Flight Condition
   - Nominal System
   - Faulty System - Stable Case

# Flight condition I

The aircraft flight condition is assumed to be a straight and constant level flight at fixed altitude of $40000$ *ft* and Mach number $0.8$.

Considering the CoG (center of gravity) position to be $(0.25, 0, 0)$ of m.a.c. (mean aerodynamic chord) and the aircraft mass $19792$ *slug*, the corresponding steady-state speed is $774$ *fps*.

The control $\delta_e$ (*rad*) is the elevator deflection.

Moreover, remember that the short-period motion involves rapid changes in the angle-of-attack $\alpha$ (*rad*) and pitch angle $\theta$ (*rad*) at roughly constant speed, with the specification that, in the small-disturbance case analysis, the evolutions of $\alpha$ and $w$ follow approximately the same shape.

# Open-loop model I

The aircraft longitudinal dynamics is described by the state vector

$$x = \begin{bmatrix} w & q \end{bmatrix}^\top ,$$

where $w$ $(ft/sec)$ and $q$ $(rad/sec)$, respectively, are the inertial velocity in the $z$-direction of $F_B$ and the pitch rate ($F_B$ represents the aircraft body-axis reference frame).

The linearized model of the system is

$$\dot{x}(t) = A_n x(t) + B_n u_n(t) ,$$

where the system and control matrices are given as follows

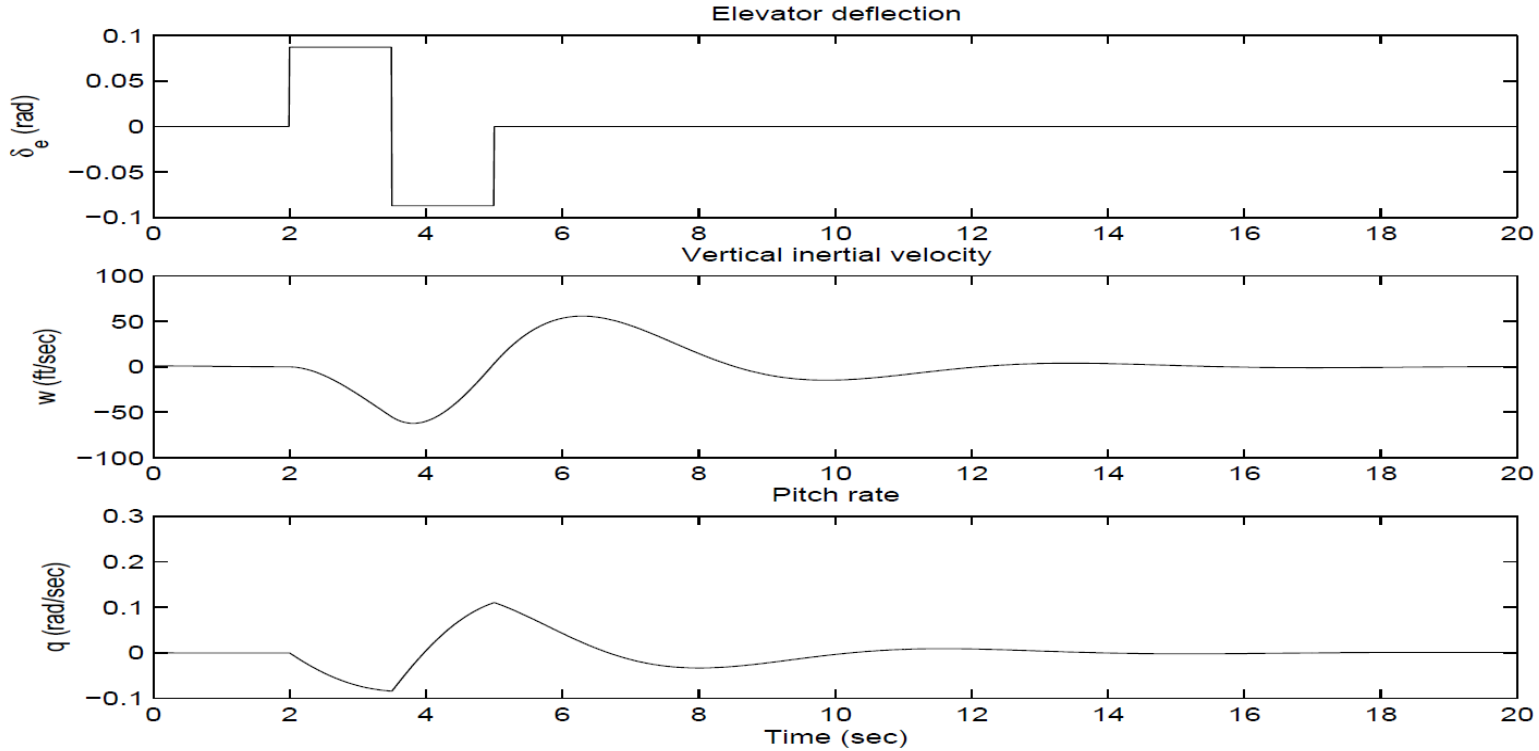
$$A_n = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix} \text{ and } B_n = \begin{bmatrix} -17.9416 \\ -1.1577 \end{bmatrix} ; \tag{15}$$

in this flight configuration, matrix $A_n$ has eigenvalues

$$\Lambda(A_n) = \{-0.3706 \pm 0.8779\jmath\} .$$

# Open-loop model II

See then the natural aircraft mode evolution in the following figure, for a 20 seconds system simulation after applying a short doublet-pulse during 3 seconds to the elevator deflection control input, where the excitation was achieved by deflecting the control column pilot input by 5 degrees (approximately 0.0872 *rad*).




Open-Loop

Behavior of B747-SP

# Closed-loop system I

Let the quadratic cost functional be

$$J[x_0, u(t)] = \int_0^{+\infty} \left[ x^\top(t) Q_n x(t) + u^\top(t) R_n u(t) \right] dt \,,$$

where the initial condition is

$$x_0 = \begin{bmatrix} 1.0000 & -0.0001 \end{bmatrix}^\top .$$

The trade-off between control amplitudes and settling time in the LQR problem is strongly influenced by the choice of the weighting matrices $Q_n$ and $R_n$; in this application, the penalty matrices are chosen $Q_n = 10^{-2} I_2$ and $R_n = 1$.

# Closed-loop system II

Then, the nominal Continuous Algebraic Riccati Equation

$$CARE_n: \quad A_n^\top X_n + X_n A_n - X_n B_n R_n^{-1} B_n^\top X_n + Q_n = 0\,,$$

with the system and control matrices (15), provides the solution

$$X_n = \begin{bmatrix} 0.0012 & 0.0630 \\ 0.0630 & 7.2792 \end{bmatrix},$$

which yields the optimal value

$$J^* = 1.2 \times 10^{-3}\,,$$

for the cost functional in (7), and gives the linear quadratic state-feedback matrix

$$K_n = \begin{bmatrix} -0.0954 & -9.5582 \end{bmatrix}. \tag{16}$$

# Closed-loop system III

In the nominal case, the closed-loop system $\hat{M}_n$ will have the closed-loop matrix $\hat{A}_n = A_n - B_n K_n$ as follows

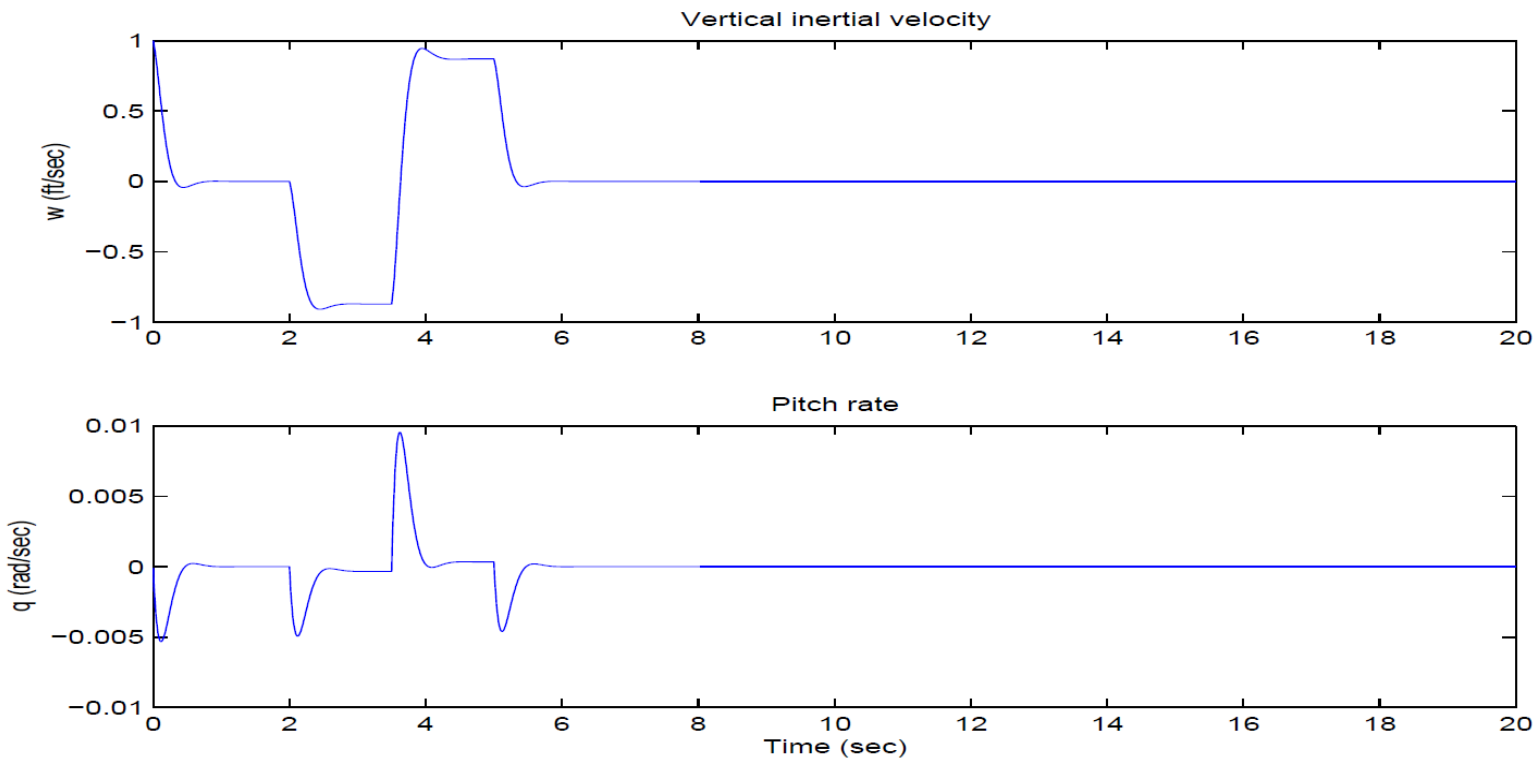
$$\hat{A}_n = \begin{bmatrix} -2.0236 & 602.5109 \\ -0.1114 & -11.4939 \end{bmatrix},$$

which is stable, with the spectrum

$$\Lambda(\hat{A}_n) = \{-6.7588 \pm 6.6857\jmath\}.$$

# Closed-loop system IV

When using (16) as nominal controller, the behavior exhibited by the closed-loop system can be seen in the following figure.




Closed-Loop

Behavior of B747-SP

# Considered faults I

The fault considered is the loss-of-efficiency of the actuator that controls the elevator deflection. Such faults are classically modeled by a multiplicative factor $\tau_f \in (0, 1]$, so that the linear state-space representation of the faulty system is described by the control matrix $B_f = (1 - \tau_f)B_n$. Obviously, $\tau_f = 0$ represents the nominal operation case and $\tau_f = 1$ stands for the complete failure case, which is the situation when the actuator does not act on the system anymore.

Still, the previous restriction to actuator faults only is too restrictive. In order to illustrate how the progressive accommodation scheme obviously works for more complex situations, consider also faults described by two coefficients, i.e., not only the control matrix $B_f = (1 - \tau_f)\, B_n$ but the faulty system matrix $A_f = (1 - \gamma_f)\, A_n$ as well, where $\gamma_f \in (0, 0.1]$.

The interpretation of parameter $\gamma_f$ may be that $A_n$ results from the linearization of a nonlinear behavior around an operating point that might well be changed by the fault occurrence or there is a systematic error in the estimation procedure ran by the diagnostic algorithm.

# Considered faults II

Consider therefore a $90\,\%$ efficiency degradation on the actuator and assume a $10\,\%$ identification error of the system matrix, that is the post-fault pair $(A_f, B_f) = [(1-\gamma_f)A_n, (1-\tau_f)B_n]$, for $\gamma_f = 0.1$ and $\tau_f = 0.9$, is given as

$$A_f = \begin{bmatrix} -0.2814 & 696.6000 \\ -0.0009 & -0.3856 \end{bmatrix} \text{ and } B_f = \begin{bmatrix} -1.7942 \\ -0.1158 \end{bmatrix};$$

then, solve the LQR problem for the penalty matrices $Q_n$ and $R_n$ introduced before.

Questions? vs. Answers!

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